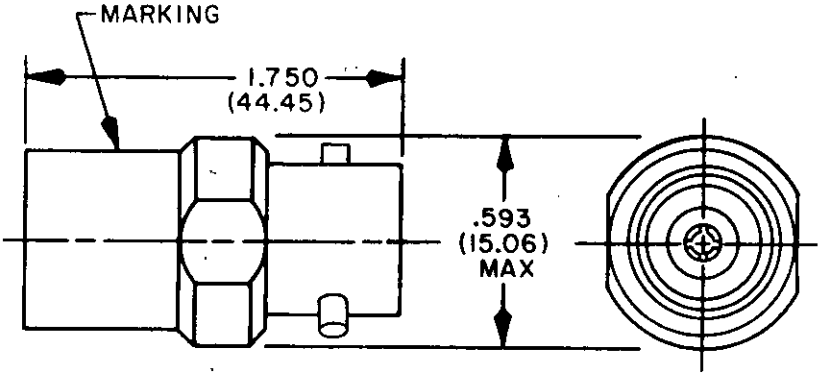
MILITARY SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY (SERIES TRB (CABLED), SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-49142.



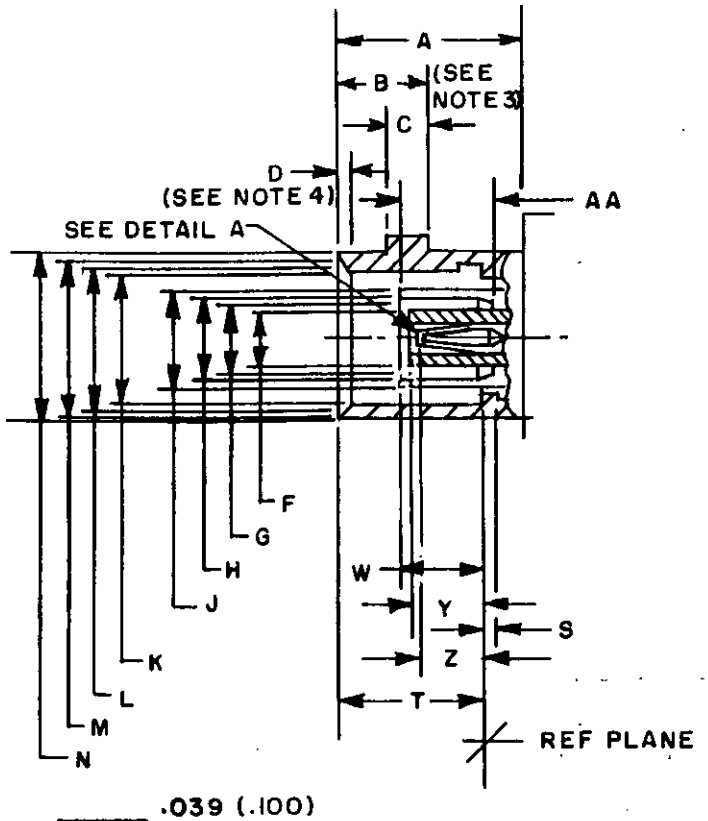
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Wrench flats are to accommodate standard wrench openings per FED-STD-H28.
5. 1.750 defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.

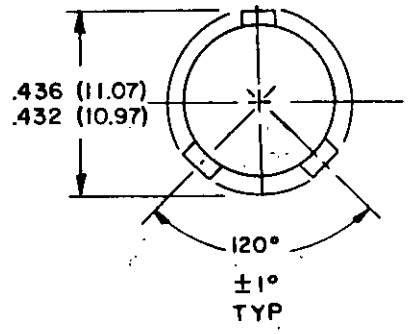
FIGURE 1. General configuration.

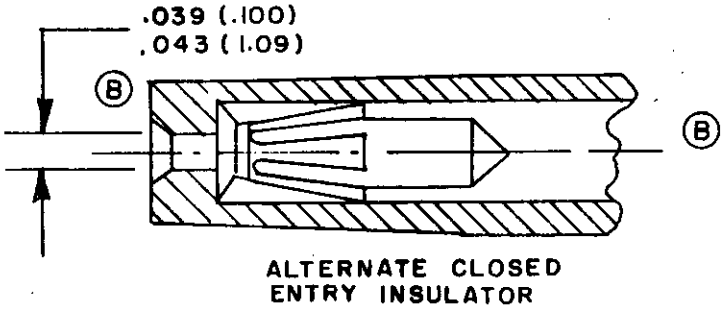
(B) denotes changes

FSC 5935

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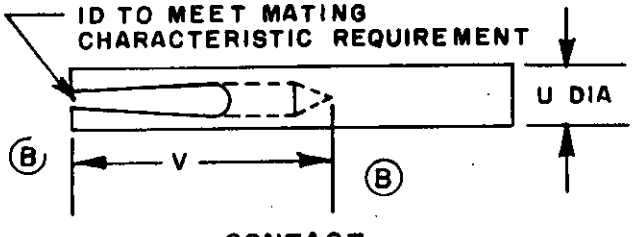






ALTERNATE CLOSED ENTRY INSULATOR

DETAIL A



CONTACT

Ltr	INCHES		MILLIMETERS	
	Min	Max	Min	Max
A	.414	---	10.52	---
(B) AA	.206	.213	5.23	5.41
B	.204	.208	5.18	5.28
C	.075	.081	1.90	2.06
D	.015	.030	0.38	0.76
F	.117	.122	2.97	3.10
G	.162	.168	4.11	4.26
H	.178	.182	4.52	4.62
J	.195	.199	4.95	5.05
K	.319	.321	8.10	8.15
L	.327	.333	8.31	8.46
M	.346	.360	8.79	9.14
N	.378	.382	9.60	9.70
S	.001	---	0.03	---
T	.327	.335	8.31	8.51
U	.062	.064	1.57	1.63
V	.213	---	5.41	---
W	.191	.213	4.85	5.41
(B) Y	---	.213	---	5.41
Z	.165	.203	4.19	5.16 (B)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. .005 (0.13 mm) flat permissible to meet dimension B.
4. Chamfer or radius.
5. Metric equivalents are in parentheses or tabulated.

FIGURE 2. Mating dimensions for socket contact termination.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Force to engage and disengage:

Longitudinal force: 4 pounds maximum.

Torque: 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See figure 2 for dimensions.

Center contact (socket):

Oversize test pin: 0.040 inch diameter, minimum (nonclosed entry contacts only).

Insertion depth: .125 inch minimum.

Number of insertions: 1.

Insertion force test:

Steel test pin finish: 16 microinches.

Insertion force: 2 pounds maximum.

Steel test pin diameter: .039 minimum +.001 inch.

Withdrawal force test:

Steel test pin diameter: .037 maximum -.001 inch.

Withdrawal force: 2 ounces minimum.

Test pin finish: 16 microinches.

Permeability: Applicable.

Hermetic seal: Not Applicable.

Leakage: Not applicable.

Insulation resistance: 5,000 megohms.

Center conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage: At sea level, 1,200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

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Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms maximum.

	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.5	0.6

Qualification: See table I.

Ⓑ TABLE I. Group qualification.

Group	Submission and qualification of any of the following dash numbers	Qualifies the following dash nos.
I	-0001 -0002 -0005	-0001 -0002 -0005
II	-0003 -0004 -0006	-0003 -0004 -0006
III	-0007 -0008	-0007 -0008

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within same series), the manufacturer may receive qualification approval for two more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design. Qualification of connectors qualifies connectors of the same body material and finish only.

Corona level:

Altitude: 70,000 feet.

Voltage: 200 V rms minimum.

RF high potential withstanding voltage:

800 V rms, between center conductor and intermediate conductor.

200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension): 40 pounds minimum.

Coupling mechanism retention force: Not applicable.

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Risetime degradation: 400 picoseconds maximum.

Ⓑ Connector durability: 500 cycles minimum at 12 cycles per minute, maximum.

Part number M49142/06-(dash number from table II).

TABLE II. Dash number and applicable cable.

*Dash number	Cable
0001	M17/134-00001
0002	M17/134-00002
0003	D3-7619-5/336
0004	D3-7619-5/338
0005	380-10045-1
0006	M17/116-RG307
0007	M17/176-00002
0008	M17/45-RG108

*Not for Naval Air Systems Command (AS) use.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3384-3)

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